Strategic Initiative for Talent Retention

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Introduction

Organizations depend on highly skilled employees as a source of competitive advantage, particularly when supported by effective talent management systems (Collings et al., 2019). According to the company's CEO, several highly capable and experienced employees have recently left the organization. Their departure may reduce innovative capacity, weaken knowledge transfer, and erode elements of the organizational culture. Attracting and retaining talented employees should therefore be a strategic priority.

Against this background, it is essential to identify which talents are leaving, understand why they are leaving, and respond quickly to retain key employees. To this end, our analysis uses a composite talent measure that captures the technical, personal, and interpersonal dimensions of employees. After identifying the talent group, we examine the drivers of voluntary turnover among these employees using exploratory data analysis and a Logistic Regression classification model. Finally, we connect our findings to suggest a mix of data-driven strategies on retaining talent for ABS Tech.

Identifying talent

Technical abilities

Human capital, including employees' knowledge and skills, is positively associated with firm performance (Crook et al., 2011). As such, the technical dimension of our formula focuses on employees' skills, knowledge and adaptability (for the full list of features, see *Figure 1*). Furthermore, the formula includes innovative contributions, which reflect employees' capacity to generate and implement novel ideas, a key source of competitive advantage and firm performance (Anderson et al., 2014).

Personal abilities

The most direct measure for personal ability is employee performance. However, performance also depends on psychological and motivational conditions (Bakker et al., 2014; Schaufeli, 2017). For example, high engagement is associated with higher task performance, organizational citizenship behavior, and proactive work behaviors, while job stress increases the risk of burnout and negatively affects performance. Finally, work–life balance reflects the alignment between job demands and personal resources, which influences burnout, work engagement, and sustainable job performance (Bakker et al., 2014).

Interpersonal abilities

The interpersonal dimension captures employees' ability to collaborate and build relationships across the organization, which is important for knowledge transfer and value creation (Nahapiet & Ghoshal, 1998). Firstly, collaboration ratings reflect how well colleagues coordinate, while networking captures the breadth and depth of employees' social connections within the organization (Borgatti & Foster, 2003). Secondly, colleague assessments and teamwork provide insight into how employees are perceived by peers and their effectiveness in team-based contexts.

The formula

The aforementioned dimensions were combined into one coherent talent formula. We decided to stick to a highly interpretable, rather than a complicated, model for ease of interpretability. Each component in the formula contributes proportionally to the total talent score with weights depicted in *Figure 1*. Note that each of the variables in the formula was first standardized with z-score scaling to ensure that uneven magnitudes across different features do not bias the final score.

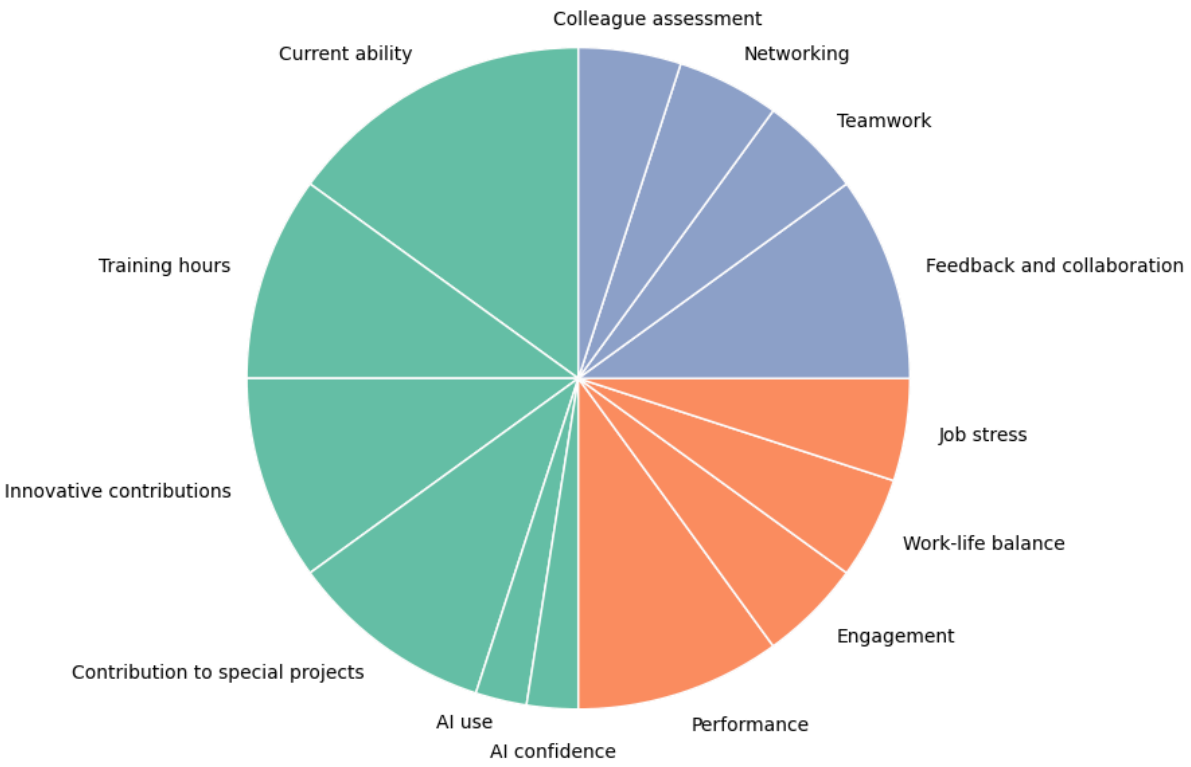

Figure 1: The 3 dimensions of talent formula. Green represents technical (50%), orange personal (25%) and purple interpersonal (25%) aspects.

Talent distribution

After assigning a talent score to each employee, we calculated the mean talent score μ and standard deviation σ. An employee was considered talent if their score was more than one standard deviation away from the mean (score ≥ μ+σ). As such, 258 employees were highlighted as non-talents, whereas the other 53 employees were considered as talent.

Note how the talent is distributed per department. *Figure 2* shows that while the production department employs the largest percentage of the total workforce, it has a relatively small proportion of talent. In contrast, the IT/IS department contains the highest concentration of talented employees relative to its size.

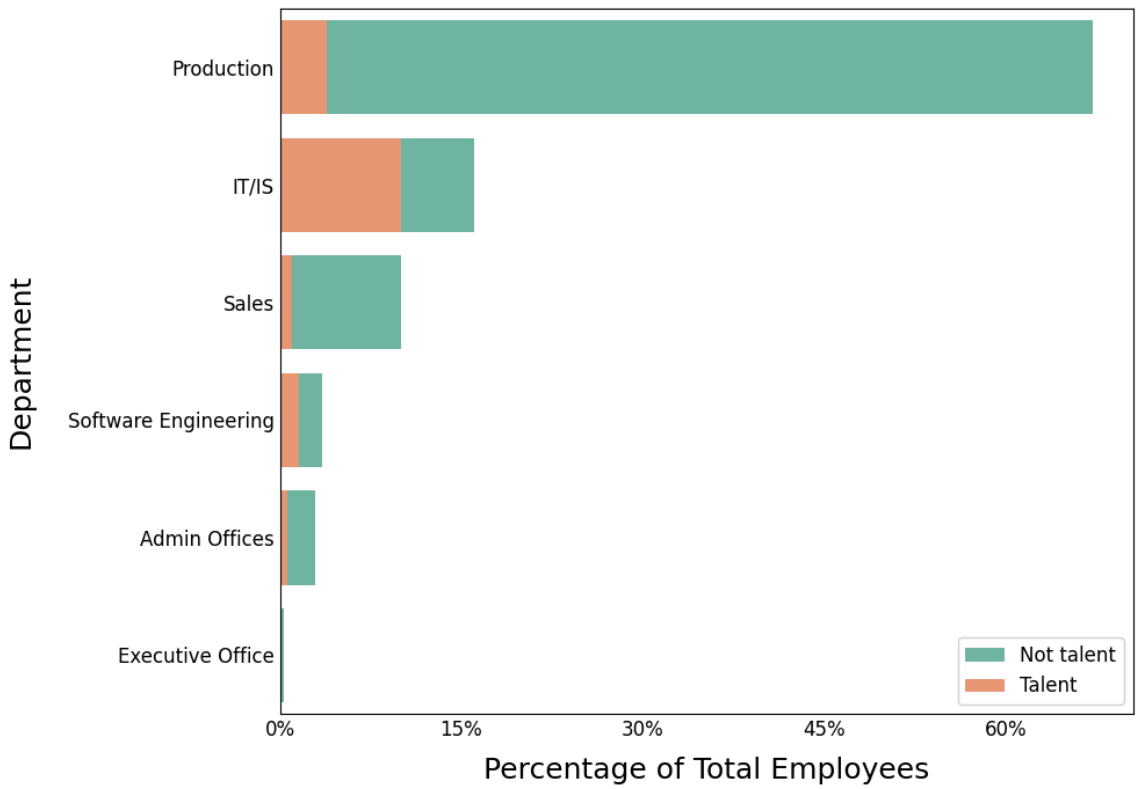

Figure 2: The distribution of non-talent and talent employees per department.

Turnover analysis

Background research

The strategic importance of talent management has grown considerably in recent years. It has been shown that organizations investing in structured talent management consistently outperform peers on organizational performance (Collings et al., 2019). Prior research suggests that employees are most likely to quit when low job satisfaction, low organizational commitment, and attractive external opportunities coincide (Hom et al., 2017). Key predictors of turnover include limited career advancement, work overload, and poor leader-member relationships, while learning opportunities and a supportive environment tend to mitigate these risks (Allen et al., 2010; Halbesleben & Wheeler, 2008). For talents specifically, these risks are amplified, as they are more likely to leave when their skills are underutilized or under-recognized (Dries, 2013; De Boeck et al., 2018). The effects of hybrid and remote work since COVID-19 remain ambiguous. On the one hand, it weakens organizational connectedness due to social isolation of employees, but it also offers benefits such as increased job performance, job satisfaction and lesser turnover intentions (Contreras et al., 2020).

General trend

The data reveals a steady increase in the voluntary turnover rate since 2014 where it peaked around 9% in 2019-2020 (see *Figure 3*). Voluntary turnover rate has since declined. Industry benchmarks indicate that technology firms typically experience an annual employee turnover rate of around 13%, which would indicate that ABS Tech is well below the average (TecBeats, 2023). However, we do believe that there is plenty of improvement to further tackle the issue of voluntary turnover.

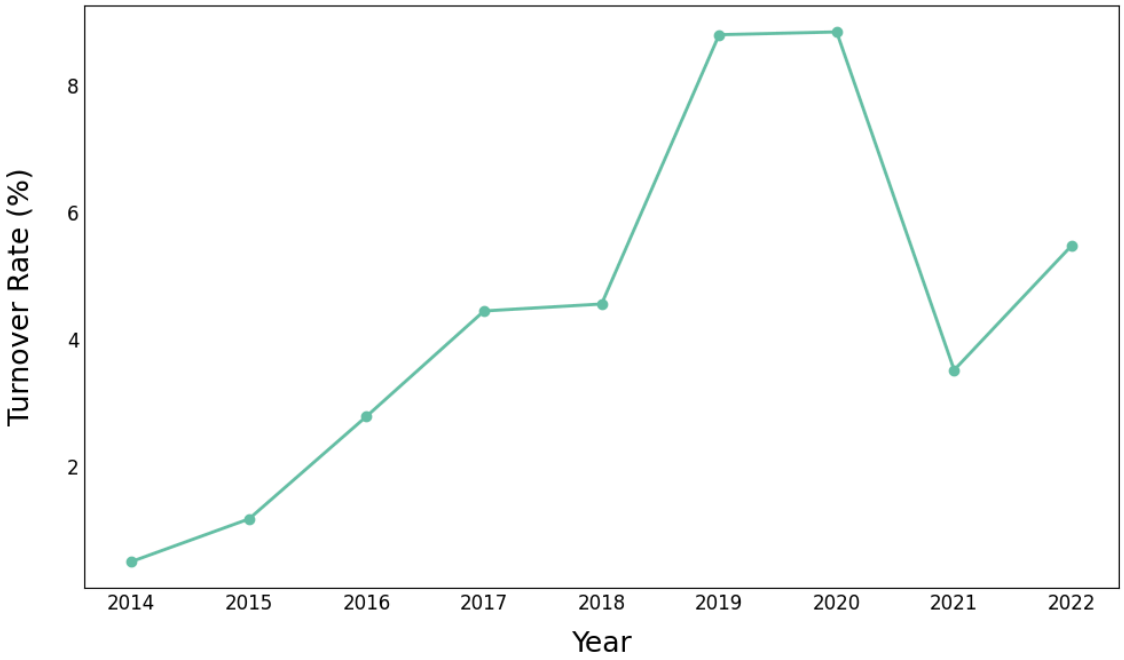

Figure 3: Turnover rate trend for ABS Tech across eight years (2014-2022).

Self-reporter reasons

As seen in *Figure 4*, employees primarily leave due to finding another position, being unhappy, or seeking better pay/career opportunities. However, talents follow a different trend, citing another position, more money, career changes, and medical issues as their primary reasons for departure.

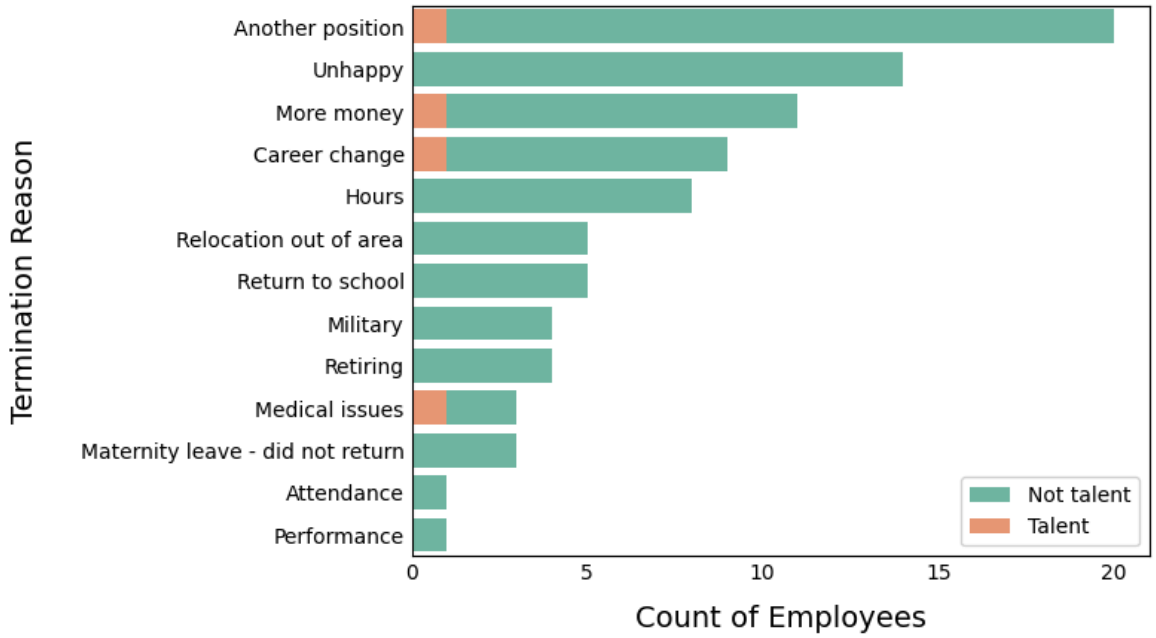

Figure 4: Self-reported reasons for voluntarily leaving the company across non-talent and talent employees.

Machine learning model

Significant variables

An essential step in data preprocessing is to highlight which features are meaningful. Therefore, we have conducted statistical tests for all features presented in the dataset that could be used for predicting whether a person is active or voluntarily terminated. We used Welch t-test statistics for numerical features and Chi-Square for categorical, and the p-values that were outputted by the tests indicated that almost all features were significant. For full results, see *Table 1* in the Appendix.

The model

To understand why employees are leaving, we built a Logistic Regression model, which was trained to predict EmploymentStatus using statistically significant features. The process included cross-validation for hyperparameter tuning and leaving 20 random observations for evaluation after the final model is built. In *Figure 5*, the main features used by the model can be seen.

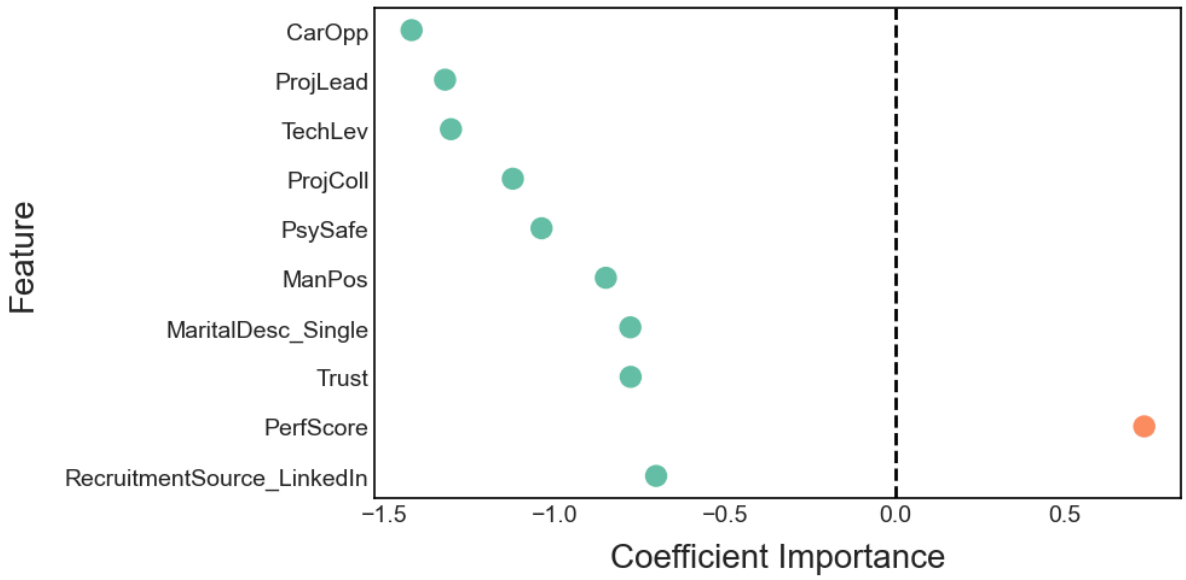

Figure 5: Positive coefficient implies positive correlation with risk of leaving, while negative implies negative (e.g. increase in CarOpp results in decrease in risk). The absolute value of coefficient importance indicates how essential regressor is in predicting the probability of leaving.

For evaluation, we used the model to predict the probability of termination for the 20 employees not included in the training set. Then we compared the model predictions to the true value of their employment status. The results are presented in *Figure 6,* showcasing great capabilities of the model

to predict high-risk and low-risk employees. Even though the model achieved perfect prediction accuracy, it should be stated that in reality, the model will most certainly not generalize this well.

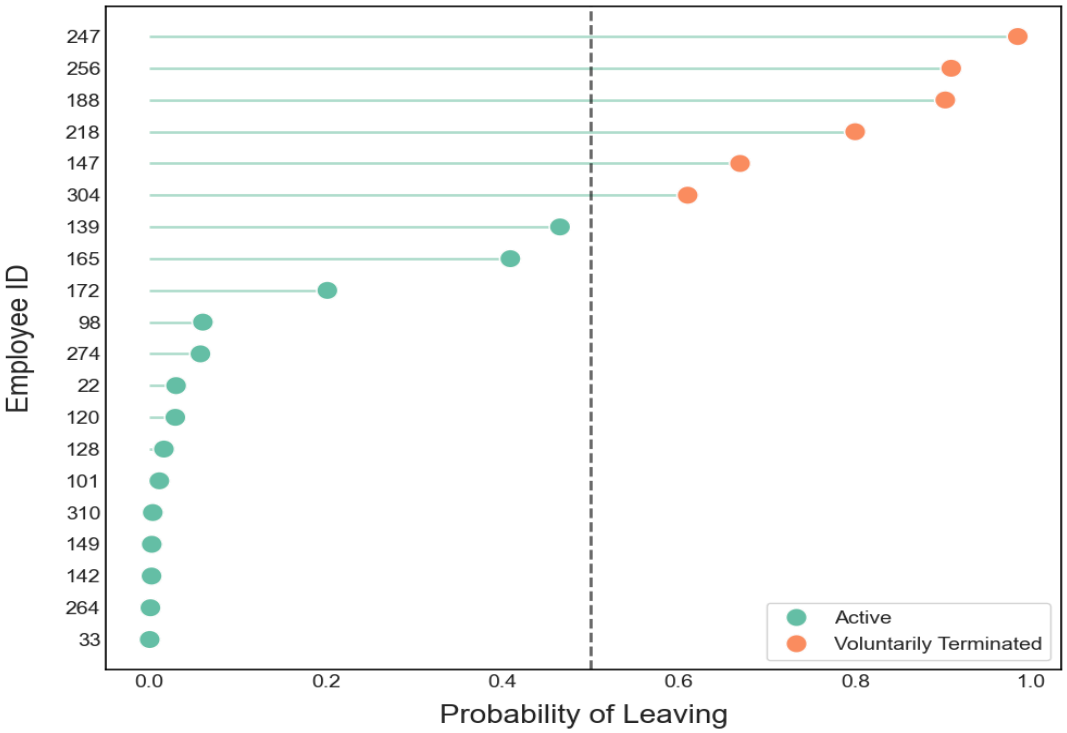

Figure 6: Testing model predictions against the actual employment status for 20 employees. Prediction threshold set at 50%.

Finally, *Figure 7* presents the probability of turnover for current talent and showcases that 6 employees are at higher risk than others, indicating that action must be taken.

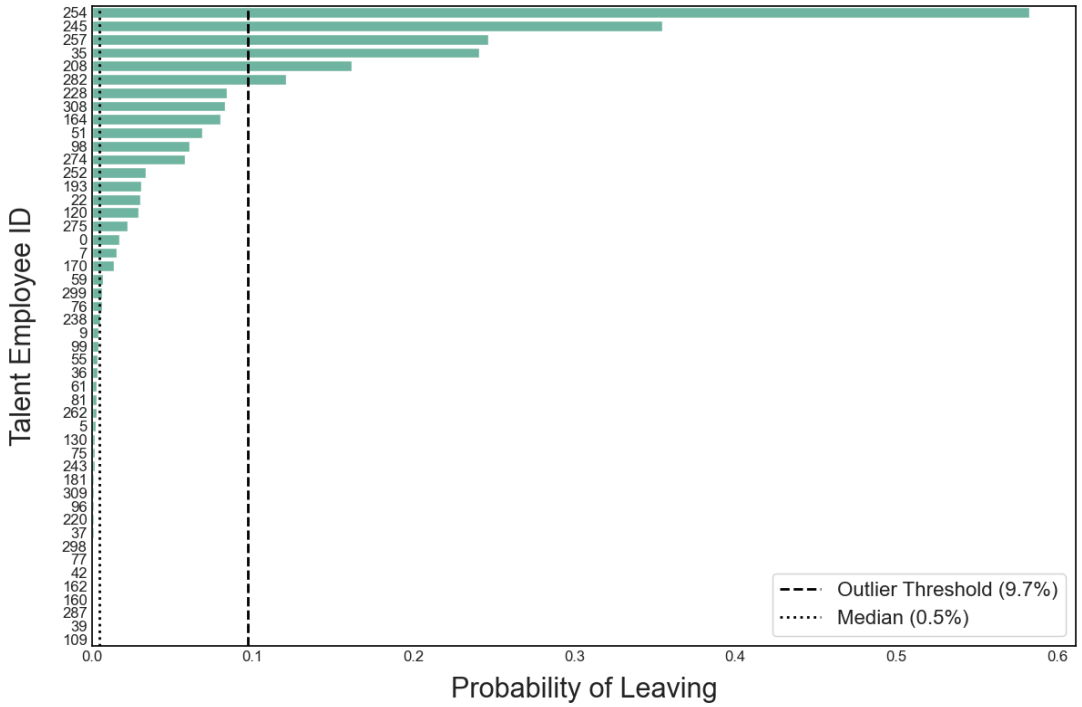

Figure 7: Current active talent and their probability of leaving. The threshold was determined by using the empirical Q2 + 1.5IQR rule.

Features of high-risk talents

Figure 8 presents greatest differences in average feature scores of current active talents versus the 6 high-risk talents. The results show that these six employees are feeling psychologically unsafe, experience a lot of stress, have lower satisfaction of working and are absent more often, relative to other talents.

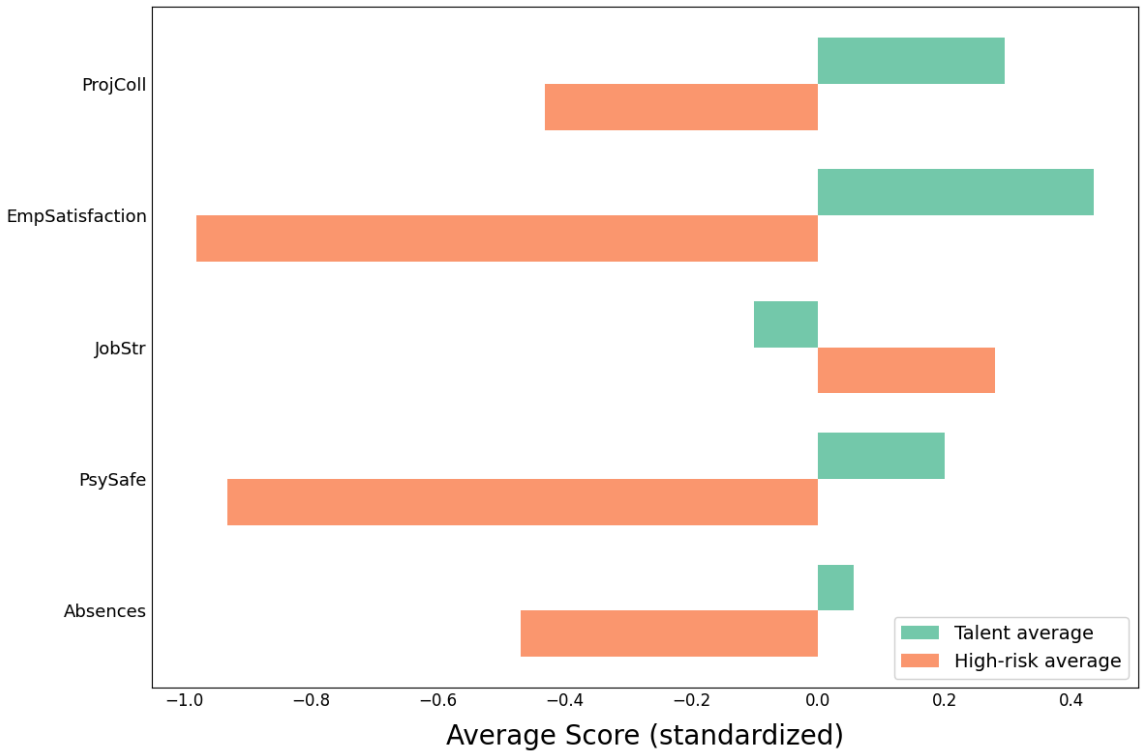

Figure 8: Greatest average differences in features of all active talents vs. high-risk of leaving talents.

In conclusion, evidence suggests that voluntary turnover at ABS Tech is driven by factors at the personal, relational, and organizational levels. At the personal level, factors such as high stress, and reduced work satisfaction increase the likelihood that employees will leave. At the relational level, low psychological safety, lack of project leads, collaboration and trust further enforce this trend. Finally, at the organizational level, limited perceived career opportunities and dissatisfaction with the work environment add to this trend.

Recommendations

Predictive dashboard (big bet)

We can translate our analysis into a live dashboard that assigns each employee a predicted probability of turnover. The goal of the dashboard is to make talent retention easier by monitoring which employees are at risk. Having such information is high-impact as it shows which employees should be prioritized for retention. Nonetheless, this is a high-effort solution as making the models and data resilient for long-term usage is a non-straightforward job.

Clear career paths (quick win)

Analysis shows that lack of career opportunities is a strong driver of voluntary termination. We recommend introducing clear career paths per department, specifying required skills and concrete developmental activities. Managers should discuss career opportunities (semi-)annually with employees during performance reviews. We believe that this solution is low-effort but high-impact as it directly tackles one of the main turnover drivers.

Better mental health (quick win)

Analysis showed that six talents are high-risk and struggling with low psychological safety and satisfaction. Our suggestion is to hire a therapist who would help to mentally manage workload and bad thoughts. The intervention is high-impact as it would reduce turnover risk and increase employee well-being. It is also low-effort as hiring a therapist is relatively straightforward.

Business value

Recommendation	Annual costs	Description
Live Dashboard	**70 000,00 €**	*Construction, maintenance, salary for data engineer/software developer*
Career Development Workshops	**15 000,00 €**	*Administrative hours for mapping and training managers*
Therapist Appointments	**60 000,00 €**	*Salary/contract for a dedicated professional*
Total:	**145 000,00 €**	

Benefit	Annual monetized value	Description
Retained Productivity Revenue	**212 099,50 €**	*= 6 high-risk of leaving employees' salaries / 2;* *New employees take 6 months to gain full productivity, then retaining 6 employees saves 6 months out of a year of full productivity (Behbahani, 2022)*
Mental Health Improvement	**55 145,87 €**	*= 6 high-risk of leaving employees' salaries * 13%;* *Based on Oxford research, mental health improvements on average increase productivity/efficiency of employee by 12-13% (De Neve et al., 2019)*
Upskilling for Margin	**28 000,00 €**	*= managers' salary (~70 000 Eur.) * 20% * 2 employees;* *Internally moving employees to senior costs 0Eur., while hiring from outside costs a headhunter's fee of around 20%. Assuming that 2 employees from 4 high-risk of leaving will become managers this year, this is the value created by retaining the employees and making them managers (Bidwell, 2012)*
Other benefits	**> 30 000,00 €**	*If employees feel more valuable, feel better and understand their career path options, they might innovate more, suggest more ideas, engage and interact more*
Total:	**295 245,37 €**	
ROI	**103,62%**	
Net Business Value	**150 245,37 €**	

Table 1: Annual Costs and Monetized Value for the presented suggestions

Table 1 presents the net business value that would be created if the above mentioned recommendations are implemented. The total costs are estimated at ~145,000 Eur.; however, if the assumption holds that all six high-risk employees are retained, then the created business value would reach a total of ~295,000 Eur.

References

Allen, D. G., Bryant, P. C., & Vardaman, J. M. (2010). Retaining talent: Replacing misconceptions with evidence-based strategies. *Academy of Management Perspectives, 24*(2), 48–64. https://doi.org/10.5465/amp.24.2.48

Anderson, N., Potočnik, K., & Zhou, J. (2014). Innovation and creativity in organizations: A state-of-the-science review, prospective commentary, and guiding framework. *Journal of Management, 40*(5), 1297–1333. https://doi.org/10.1177/0149206314527128

Bakker, A. B., Demerouti, E., & Sanz-Vergel, A. I. (2014). Burnout and work engagement: The JD–R approach. *Annual Review of Organizational Psychology and Organizational Behavior, 1*(1), 389–411. https://doi.org/10.1146/annurev-orgpsych-031413-091235

Behbahani, N (2022). New Hires: Time to Productivity vs. Time to Hire. Retrieved March 28, 2026, from https://www.makingbusinessmatter.co.uk/time-to-productivity/

Bidwell, M. (2012). Why External Hires Get Paid More, and Perform Worse, than Internal Staff. Retrieved March 28, 2026, from https://knowledge.wharton.upenn.edu/article/why-external-hires-get-paid-more-and-perform-worse-than-internal-staff/

Borgatti, S. P., & Foster, P. C. (2003). The network paradigm in organizational research: A review and typology. *Journal of Management, 29*(6), 991–1013. https://doi.org/10.1016/S0149-2063(03)00087-4

Collings, D. G., Mellahi, K., & Cascio, W. F. (2019). Global talent management and performance in multinational enterprises: A multilevel perspective. *Journal of Management, 45*(2), 540–566. https://doi.org/10.1177/0149206318757018

Contreras, F., Baykal, E., & Abid, G. (2020). E-Leadership and teleworking in times of COVID-19 and Beyond: What we know and where do we go. *Frontiers in Psychology*, *11*, 590271. https://doi.org/10.3389/fpsyg.2020.590271

Crook, T. R., Todd, S. Y., Combs, J. G., Woehr, D. J., & Ketchen, D. J. (2011). Does human capital matter? A meta-analysis of the relationship between human capital and firm performance. *Journal of Applied Psychology, 96*(3), 443–456. https://doi.org/10.1037/a0022147

De Boeck, G., Meyers, M. C., & Dries, N. (2018). Employee reactions to talent management: Assumptions versus evidence. *Journal of Organizational Behavior, 39*, 199–213. https://doi.org/10.1002/job.2254

De Neve J., Ward G. & Bellet C. (2019). Happy workers are 13% more productive. Retrieved March 28, 2026, from https://www.ox.ac.uk/news/2019-10-24-happy-workers-are-13-more-productive

Dries, N. (2013). The psychology of talent management: A review and research agenda. *Human Resource Management Review, 23*(4), 272–285. https://doi.org/10.1016/j.hrmr.2013.05.001

Halbesleben, J. R. B., & Wheeler, A. R. (2008). The relative roles of engagement and embeddedness in predicting job performance and intention to leave. *Work & Stress, 22*(3), 242–256. https://doi.org/10.1080/02678370802383962

Hom, P. W., Lee, T. W., Shaw, J. D., & Hausknecht, J. P. (2017). One hundred years of employee turnover theory and research. *Journal of Applied Psychology, 102*(3), 530–545. https://doi.org/10.1037/apl0000103

Nahapiet, J., & Ghoshal, S. (1998). Social capital, intellectual capital, and the organizational advantage. *Academy of Management Review, 23*(2), 242–266. https://doi.org/10.5465/amr.1998.533225

Schaufeli, W. B. (2017). Applying the job demands-resources model. *Organizational Dynamics, 46*(2), 120–132. https://doi.org/10.1016/j.orgdyn.2017.04.008

TecBeats. (2021). Attrition rate in tech: What is normal for the tech industry? Retrieved March 28, 2026, from https://tecbeats.com/attrition-rate-in-tech

Appendix

Feature	Type	Test used	P-value	Significance	Effect	Note
CarOpp	Numeric	Welch t-test	1.87E-15	Significant	Higher in Active	Mean Diff: 0.92
OrgIden	Numeric	Welch t-test	2.51E-12	Significant	Higher in Active	Mean Diff: 0.85
ProjSelf	Numeric	Welch t-test	1.44E-11	Significant	Higher in Active	Mean Diff: 0.87
Trust	Numeric	Welch t-test	1.47E-11	Significant	Higher in Active	Mean Diff: 0.86
ProjColl	Numeric	Welch t-test	4.38E-08	Significant	Higher in Active	Mean Diff: 0.70
TeamIden	Numeric	Welch t-test	1.07E-07	Significant	Higher in Active	Mean Diff: 0.67
ProjLead	Numeric	Welch t-test	1.42E-07	Significant	Higher in Active	Mean Diff: 0.65
RecruitmentSource	Categorical	Chi-Square	2.47E-07	Significant	N/A (Group Association)	Degrees of Freedom: 8
AIConf	Numeric	Welch t-test	3.35E-07	Significant	Higher in Active	Mean Diff: 0.59
PsySafe	Numeric	Welch t-test	6.59E-07	Significant	Higher in Active	Mean Diff: 0.62
Feedback	Numeric	Welch t-test	3.91E-06	Significant	Higher in Active	Mean Diff: 0.59
Network	Numeric	Welch t-test	7.56E-06	Significant	Higher in Active	Mean Diff: 0.55
AIUse	Numeric	Welch t-test	9.71E-05	Significant	Higher in Active	Mean Diff: 0.48
SpecialProjectsCount	Numeric	Welch t-test	1.13E-04	Significant	Higher in Active	Mean Diff: 0.41
TechLev	Categorical	Chi-Square	5.61E-04	Significant	N/A (Group Association)	Degrees of Freedom: 8
MaritalDesc	Categorical	Chi-Square	2.69E-03	Significant	N/A (Group Association)	Degrees of Freedom: 4
TrainHours	Numeric	Welch t-test	4.08E-03	Significant	Higher in Active	Mean Diff: 0.36
ManPos	Categorical	Chi-Square	7.82E-03	Significant	N/A (Group Association)	Degrees of Freedom: 1
Salary	Numeric	Welch t-test	1.14E-02	Significant	Higher in Active	Mean Diff: 6744.83
InnoCont	Categorical	Chi-Square	3.67E-02	Significant	N/A (Group Association)	Degrees of Freedom: 4
Remote	Categorical	Chi-Square	1.28E-01	Not Significant	N/A (Group Association)	Degrees of Freedom: 2
Absences	Numeric	Welch t-test	1.31E-01	Not Significant	Higher in Terminated	Mean Diff: -0.19
JobStr	Numeric	Welch t-test	3.27E-01	Not Significant	Higher in Active	Mean Diff: 0.13
PerfScore	Numeric	Welch t-test	3.89E-01	Not Significant	Higher in Active	Mean Diff: 0.10
DaysLateLast30	Numeric	Welch t-test	4.54E-01	Not Significant	Higher in Terminated	Mean Diff: -0.09
EngagementSurvey	Numeric	Welch t-test	5.18E-01	Not Significant	Higher in Terminated	Mean Diff: -0.08
RaceDesc	Categorical	Chi-Square	8.12E-01	Not Significant	N/A (Group Association)	Degrees of Freedom: 5
WLF	Numeric	Welch t-test	8.80E-01	Not Significant	Higher in Active	Mean Diff: 0.02
EmpSatisfaction	Numeric	Welch t-test	9.72E-01	Not Significant	Higher in Terminated	Mean Diff: -0.00

Table 2: Statistical analysis on feature significance between voluntarily terminated and active employees.

Github repo with full analysis can be accessed via the following:
https://github.com/DovydasVe/HR_analytics_protecting_talent